

# Reflect Points Across Axes

Lesson 9-7

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*$(3, 2)$  reflected over the  $y$ -axis becomes  $(-3, 2)$  —  $x$  changes sign,  $y$  stays*

### Reflection

A flipped image of a point on the other side of a line, the same distance away.

*$(4, 3) \rightarrow (4, -3)$ : the  $y$  flips from  $+3$  to  $-3$ , like folding the paper along the horizontal line*

### x-axis

The line that goes across the grid. Flipping over it changes the  $y$  sign.

*$(4, 3) \rightarrow (-4, 3)$ : the  $x$  flips from  $+4$  to  $-4$ , like folding the paper along the vertical line*

### y-axis

The line that goes up the grid. Flipping over it changes the  $x$  sign.

*A butterfly's wings are symmetric — fold it down the middle and both sides match*

### Symmetry

When a shape looks the same on both sides of a line.

*..., -3, -2, -1, 0, 1, 2, 3, ...*

### Integer

Whole numbers and their opposites, like  $-2$ ,  $-1$ ,  $0$ ,  $1$ ,  $2$ .

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can reflect points across the x-axis and y-axis on the coordinate plane.**

#### 1 Notice

Captain Vega discovers that her treasure map has a magic mirror trick!

#### 2 Key idea

I can reflect points across the x-axis and y-axis on the coordinate plane.

#### 3 Words I'll use

Reflection

x-axis

y-axis

Symmetry

Integer



#### Think About It

- When  $(2, 4)$  reflects over the x-axis to  $(2, -4)$ , which coordinate changed?
- When  $(5, 3)$  reflects over the y-axis to  $(-5, 3)$ , which coordinate changed?
- What stays the same in each reflection?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**When you reflect the point  $(3, 2)$  over the  $y$ -axis, which coordinate changes its sign and which stays the same? Why?**

#### Level 1 support

**Start here:** Reflecting over the  $y$ -axis changes the \_\_\_\_ coordinate because \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**The \_\_\_\_ coordinate changes sign.**

*La coordenada \_\_\_\_ cambia de signo.*

**The \_\_\_\_ coordinate stays the same.**

*La coordenada \_\_\_\_ se queda igual.*

**Word bank:**

reflection

$x$ -axis

$y$ -axis

ordered pair

coordinate plane

**Listen for:** Student states reflecting over the  $y$ -axis flips the sign of the  $x$ -coordinate while the  $y$ -coordinate stays the same, so  $(3, 2)$  becomes  $(-3, 2)$ .

#### Level 2

**Push students to explain why the reflected point is the same distance from the  $y$ -axis as the original.**

- The reflected point is the same distance from the axis because \_\_\_\_.
- A reflection is like a \_\_\_\_ over the line.

EXPLORE

**Reflect  $(4, -5)$  over the  $x$ -axis. Talk through how the coordinates change and where the new point lands.**

Level 1 support

**Start here:** Reflecting over the  $x$ -axis changes the \_\_\_\_ coordinate, so the point becomes \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**Over the  $x$ -axis, the \_\_\_\_ coordinate flips sign.**

*Sobre el eje  $x$ , la coordenada \_\_\_\_ cambia de signo.*

**So  $(4, -5)$  becomes \_\_\_\_.**

*Entonces  $(4, -5)$  se convierte en \_\_\_\_.*

**Word bank:** reflection x-axis y-axis ordered pair coordinate plane

**Listen for:** Student flips the sign of the  $y$ -coordinate, so  $(4, -5)$  becomes  $(4, 5)$ , and explains the  $x$  stays the same.

Level 2

**Ask students to predict where  $(4, -5)$  lands if reflected over the  $y$ -axis instead, and compare the two results.**

- Reflected over the  $y$ -axis,  $(4, -5)$  becomes \_\_\_\_.
- The two reflections are different because \_\_\_\_.

CONNECT

**How is reflecting a point over an axis like looking at it in a mirror placed on that axis?**

Level 1 support

**Start here:** A reflection is like a mirror because \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**The axis acts like the \_\_\_\_.**

*El eje actua como el \_\_\_\_.*

**The image is the same distance away but on the \_\_\_\_ side.**

*La imagen esta a la misma distancia pero del \_\_\_\_ lado.*

**Word bank:** reflection x-axis y-axis coordinate plane ordered pair

**Listen for:** Student connects the axis to a mirror line and the reflected point to a mirror image, equal distance on the opposite side.

Level 2

**Push students to generalize a rule: what changes when you reflect over the x-axis versus the y-axis?**

- Reflecting over the x-axis changes \_\_\_\_, while reflecting over the y-axis changes \_\_\_\_.
- A rule I can always use is \_\_\_\_.

REFLECT

A classmate reflected  $(2, 3)$  over the  $x$ -axis and got  $(-2, 3)$ . What did they do wrong, and what is the correct image?

Level 1 support

**Start here:** The mistake was changing the \_\_\_\_ coordinate, but the correct image is \_\_\_\_.



Need a hint?

Sentence starters (optional)

Over the  $x$ -axis, they should have changed the \_\_\_\_ coordinate.

*Sobre el eje  $x$ , debieron cambiar la coordenada \_\_\_\_.*

The correct image is \_\_\_\_.

*La imagen correcta es \_\_\_\_.*

**Word bank:** reflection x-axis y-axis ordered pair coordinate plane

**Listen for:** Student identifies they flipped  $x$  instead of  $y$ ; the correct image over the  $x$ -axis is  $(2, -3)$ .

Level 2

Ask students to critique how to remember which coordinate changes for each axis.

- To remember which coordinate changes, I think \_\_\_\_.
- Reflecting over the  $x$ -axis affects the \_\_\_\_ value because \_\_\_\_.

## Guided Examples

### Example 1

**What is the reflection of  $(4, -3)$  over the x-axis?**

**Solution:** Reflecting over the x-axis changes the sign of the y-coordinate. The x stays the same (4), and the y changes from -3 to 3. The answer is  $(4, 3)$ .

**Answer:** B.  $(4, 3)$

### Example 2

**What is the reflection of  $(-2, 6)$  over the y-axis?**

**Solution:** Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2, and y stays 6. The answer is  $(2, 6)$ .

**Answer:** A.  $(2, 6)$

### Example 3

**Which point is on the y-axis?**

**Solution:** Points on the y-axis have x-coordinate = 0.  $(0, 5)$  is on the y-axis.

**Answer:** A.  $(0, 5)$

## Write About the Math

### The Writing Revolution

I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

#### 1. Kernel Sentence subject + verb

**Model:** Reflection is a flipped image of a point on the other side of a line, the same distance away.  
*Reflexión es una imagen volteada de un punto al otro lado de una línea, a la misma distancia.*

**Write a kernel sentence about reflection. Use a subject and a verb.**

*Escribe una oración base sobre reflexión. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Reflection matters in math  
*Reflexión importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Reflection matters in math because \_\_\_\_.**  
*Reflexión importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Reflection matters in math, but \_\_\_\_.**  
*Reflexión importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Reflection matters in math, so \_\_\_\_.**  
*Reflexión importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about reflection.  
*Di un hecho verdadero sobre reflection.*

**Reflection** \_\_\_\_.

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**Question**  
*Pregunta*

Ask a question about reflection.  
*Haz una pregunta sobre reflection.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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**Exclamation**  
*Exclamación*

Show excitement about reflection.  
*Muestra entusiasmo sobre reflection.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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**Command**  
*Mandato*

Tell a partner what to do with reflection.  
*Dile a un compañero qué hacer con reflection.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**The** \_\_\_\_ **coordinate changes sign.**

*La coordenada* \_\_\_\_ *cambia de signo.*

**The** \_\_\_\_ **coordinate stays the same.**

*La coordenada* \_\_\_\_ *se queda igual.*

**Over the x-axis, the** \_\_\_\_ **coordinate flips sign.**

*Sobre el eje x, la coordenada* \_\_\_\_ *cambia de signo.*

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## Try It

Solve on your own. Check the answer key when you are done.

### 1. Which point is on the y-axis?

- A. (0, 5)
- B. (5, 0)
- C. (3, 3)
- D. (-2, 4)

Show your work:

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### 2. Point A is at (-3, 5). It is reflected over the x-axis to A', then A' is reflected over the y-axis to A''. What are the coordinates of A''?

- A. (3, -5)
- B. (-3, -5)
- C. (3, 5)
- D. (-3, 5)

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A triangle has vertices at  $A(1, 2)$ ,  $B(4, 2)$ , and  $C(4, 6)$ . If you reflect the entire triangle over the  $y$ -axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?**

*Sentence starter: The reflected vertices are  $A'(\underline{\hspace{1cm}})$ ,  $B'(\underline{\hspace{1cm}})$ ,  $C'(\underline{\hspace{1cm}})$ . I found each by  $\underline{\hspace{1cm}}$ . The reflected triangle is  $\underline{\hspace{1cm}}$  compared to the original because reflections  $\underline{\hspace{1cm}}$ .*

Show your work:

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## Reflect — Exit Ticket

**What is the reflection of  $(-5, 2)$  over the  $y$ -axis?**

- A.  $(-5, -2)$
- B.  $(5, -2)$
- C.  $(5, 2)$
- D.  $(2, -5)$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A.  $(0, 5)$  — *Points on the  $y$ -axis have  $x$ -coordinate = 0.  $(0, 5)$  is on the  $y$ -axis.*
2. **Try It 2:** A.  $(3, -5)$  — *Over  $x$ -axis:  $(-3, 5) \rightarrow (-3, -5)$ . Then over  $y$ -axis:  $(-3, -5) \rightarrow (3, -5)$ . Both coordinates changed sign.*
3. **Exit Ticket:** C.  $(5, 2)$  — *Reflecting over the  $y$ -axis changes the sign of the  $x$ -coordinate.  $x$  goes from  $-5$  to  $5$ , and  $y$  stays  $2$ . The reflection is  $(5, 2)$ .*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Reflection is a flipped image of a point on the other side of a line, the same distance away.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).